Unit 4, Lesson 8: Absolute Value – Part 1

Benchmark

MA.6.NSO.1.3 Given a mathematical or real-world context, interpret the absolute value of a number as the distance from zero on a number line. Find the absolute value of rational numbers.

Learning Target

☐ I can find the absolute value of a rational number.

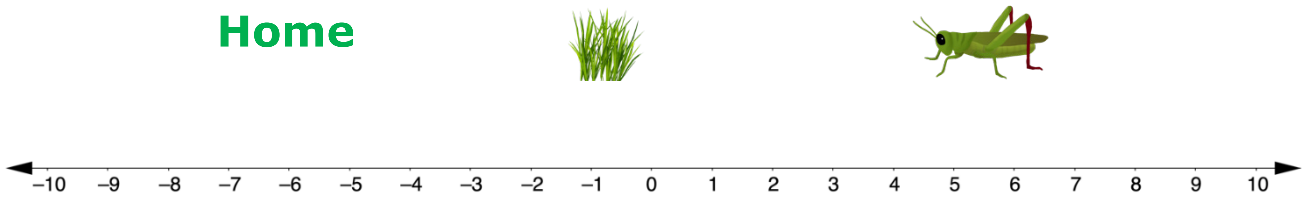
4.8.1 Warm-Up: Would You Rather?

1. Would you rather have Pile A or Pile B? Justify your reasoning.

Pile A	Pile B
14 pennies	2 pennies
7 nickels	4 nickels
2 dimes	2 dimes
1 quarter	2 quarters

4.8.2 Exploration: Jumping Bugs

A bug is jumping around on the number line. His home is located at 0. Use a cut out of the bug to answer the following questions about the bug's location.



1. The bug starts at 5 and jumps 3 units to the right.

   a. What is the bug's new location?

   b. How far away from home is the bug?

2.

   a. If the bug starts at 2 and jumps 5 units to the left, what is the bug's new location?

   b. How far away from home is the bug?

3. Mr. Rodriguez told his class to begin with the bug at home. Then the bug should jump 4 units. When Micah and Brianna did this, they discovered that their bugs were in two different locations. Discuss with your partner how this may have happened. Write your explanation or draw a picture of what could have happened.

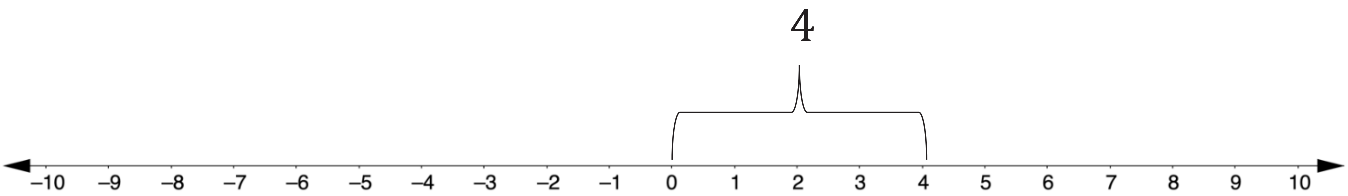
4.8.3 Guided Instruction: Absolute Value and Zero

Recall how opposite numbers relate to zero. Complete the explanation below by writing a word in each blank.

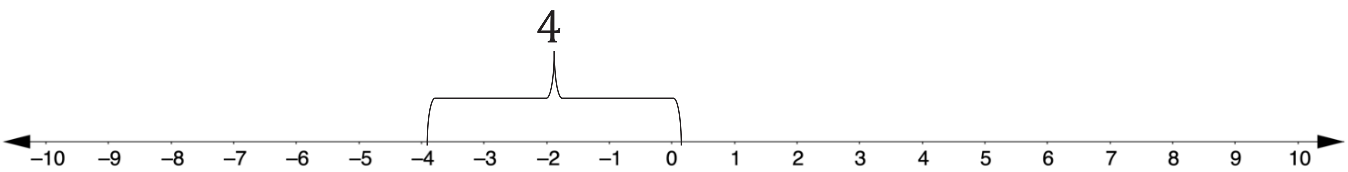
1. Pairs of opposite values are the same _______________ from zero on the number line. They are mirror images of each other, with a line of symmetry through ___________.

Absolute value is the distance a number is from zero (0) on a number line.

2. The absolute value of 4 is _______ because it is 4 units from 0 on the number line.



3. The absolute value of -4 is _______ because it is 4 units from 0 on the number line.



4. The values 4 and -4 are opposites. Each value is 4 units from zero. A number and its opposite have the same ___________________________.

4.8.4 Guided Practice: Absolute Value and Zero

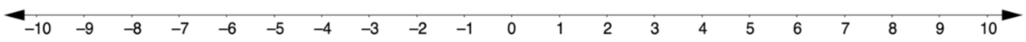
Absolute value is represented using the notation $|a|$. The notation $|a|$ is read, "the absolute value of a."

1.

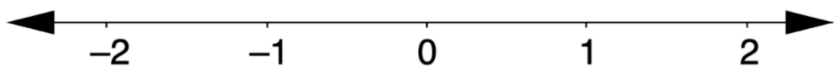
   a. The $|-8| =$ _____ because the distance from -8 to 0 is _____.

   b. What other number has the same absolute value as -8?

   c. Use the number line to show the other number that has the same absolute value as -8.



2. On the number line, plot and label all numbers with an absolute value of $\frac{3}{2}$.



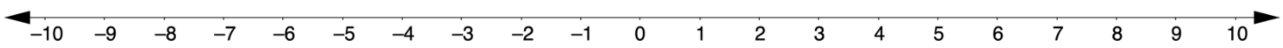
3. Hector and his partner, Salih, were given the following two problems to complete.

$|5.5| =$ _______ $\qquad\qquad$ |_______| $= 3.2$

Hector and Salih had the same answer for one of the blanks but two different answers for the other. Their teacher told them they both had the correct answers.

   a. Which problem can have more than one correct answer?

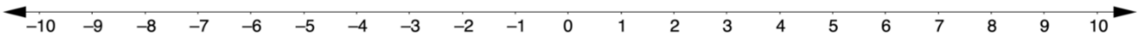
   b. Use the number line to explain how this is possible.



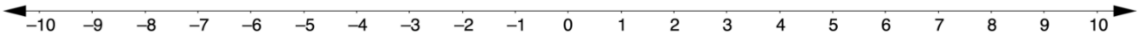
4.8.5 Your Turn!

1. Complete each statement and model it on the number line given.

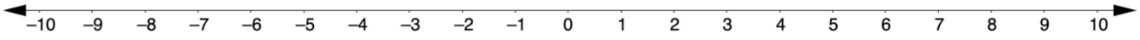
   a. $|-7.8| =$ ________



   b. $|6.25| =$ ________

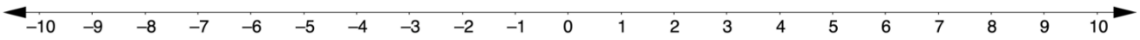


   c. $|\_\_\_\_\_| = 3\frac{1}{2}$



2. Choose any rational number between -10 and 10.

   a. Plot the number and then plot its opposite on the number line.

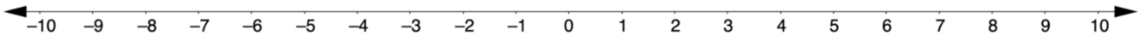


   b. What is the relationship between the absolute values of these rational numbers?

4.8.6 Wrap-Up: Ticket Out the Door

1. Complete the statement and model it on the number line.

$$\left|-\frac{10}{4}\right| = \underline{\hspace{3cm}}$$



2. What must be true about two numbers that have the same absolute value?